FACULTÉ
DE MÉDECINE
UM6SS - CASABLANCA

SEMIOLOGY OF THE PITUITARY GLAND

Pr . IOANA HRISTOV

Semiology- Endocrinology

1st semester

2026-2027

UNIVERSITÉ MOHAMMED VI
DES SCIENCES DE LA SANTÉ
CASABLANCA

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Objectives:

1. Describe clinical presentation of pituitary diseases
2. Describe the clinical presentation of hypo and hyper production of the pituitary hormones
3. Explain the laboratory investigations of the pituitary gland disorders

Headings

1. Anatomical Overview
2. Physiological Overview
3. Semiology of the anterior pituitary
4. Semiology of the posterior pituitary
5. Conclusion

PITUITARY ANATOMY

II. Pituitary :

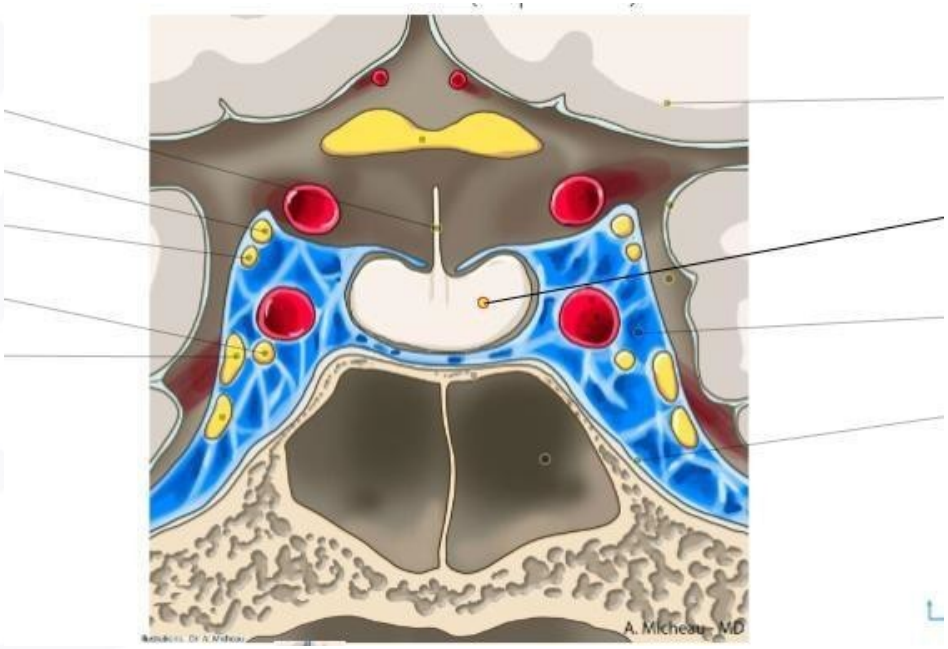
1 – Anatomy : hypophyseal fossa, the sella turcica

2 – Subdivision: anterior (adenohypophysis) and posterior (neurohypophysis or pars nervosa) lobes.

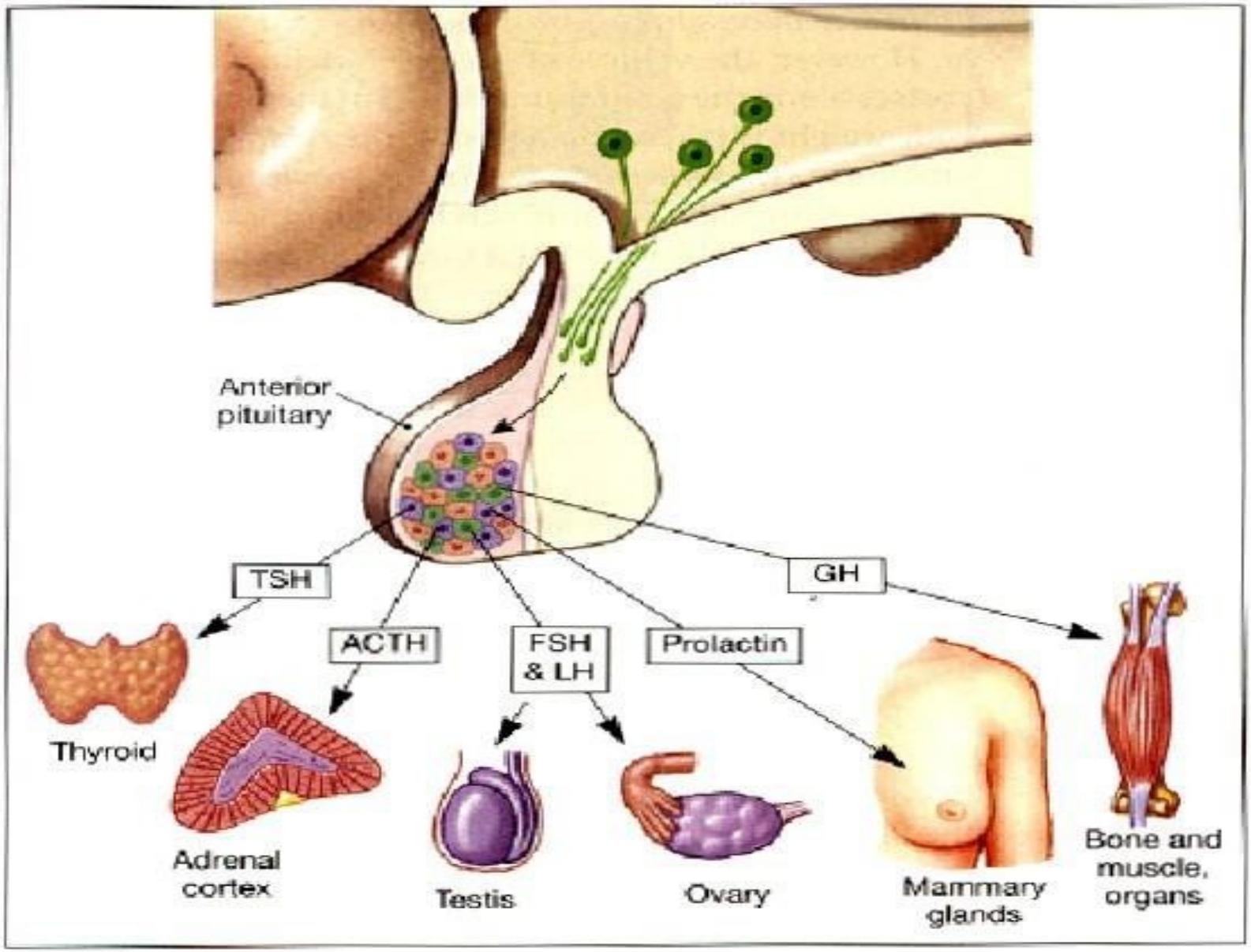
3 – Dimensions: 5×10×15 mm

4 – Region structures:

Sella turcica below
Optic chiasm above
Cavernous sinuses
on both sides



PHYSIOLOGY OF THE PITUITARY GLAND- REMINDER



SEMIOLOGY FOR PITUITARY PATHOLOGIES

- Hypersecretion FOR EACH pituitary AXIS (TROPHIC HORMONE)
- Pituitary insufficiency (ONE OR MORE AXIS)
- Mass effect

CLINICAL PRESENTATION

Hypersecretion of pituitary hormones

- **PROLACTIN HYPERSECRETION**
- **ACTH HYPERSECRETION- CUSHING DISEASE**
- **GH HYPERSECRETION**
- **TSH HYPERSECRETION**
- **FSH/LH HYPERSECRETION**

HYPERPROLACTINEMIA

Clinical Features

The clinical manifestations of PRL excess are the same regardless of the cause.

- **In women**

✓ Galactorrhea: spontaneous/induced, permanent/intermittent
✓ Menstrual disorders :Primary/secondary amenorrhea Oligomenorrhea
✓ Infertility;
✓ Low libido,

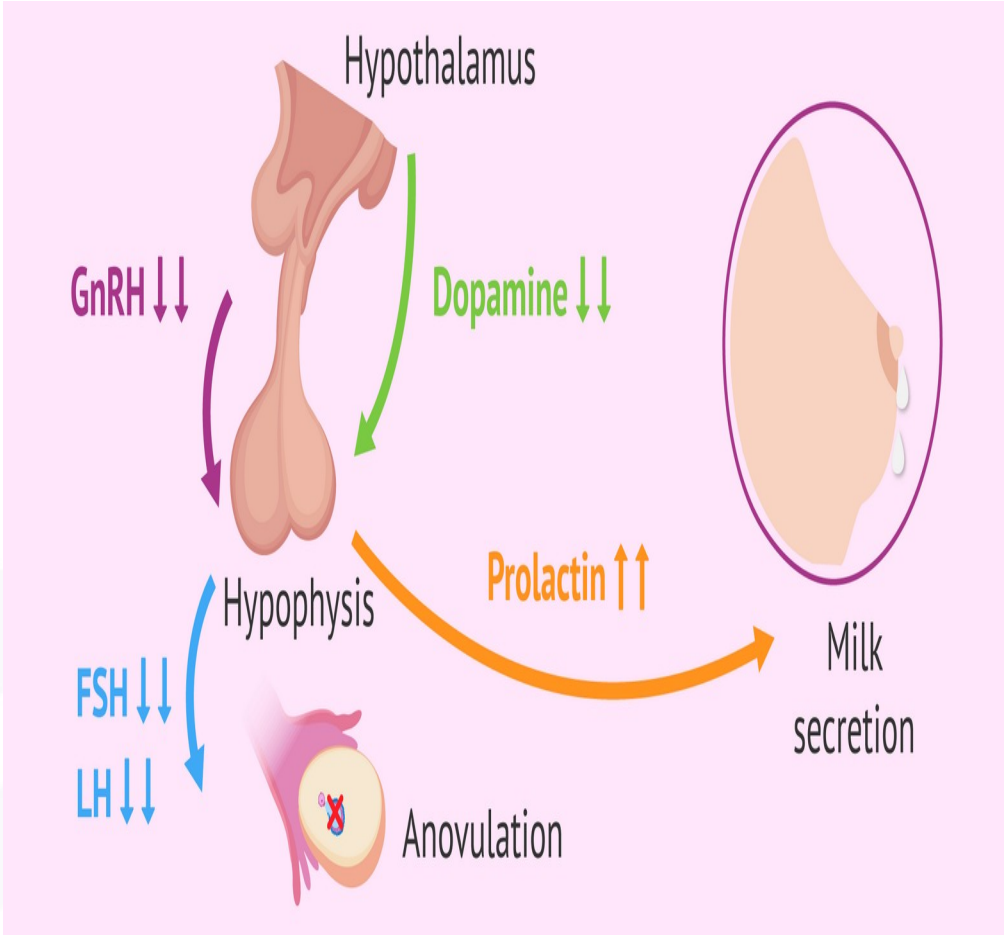
- **In men:** the usual manifestations are those of hypogonadism

✓ Low libido is the initial symptom
✓ Erectile dysfunction
✓ Gynecomastia
✓ Infertility

Mass effects (macroadenomas only):

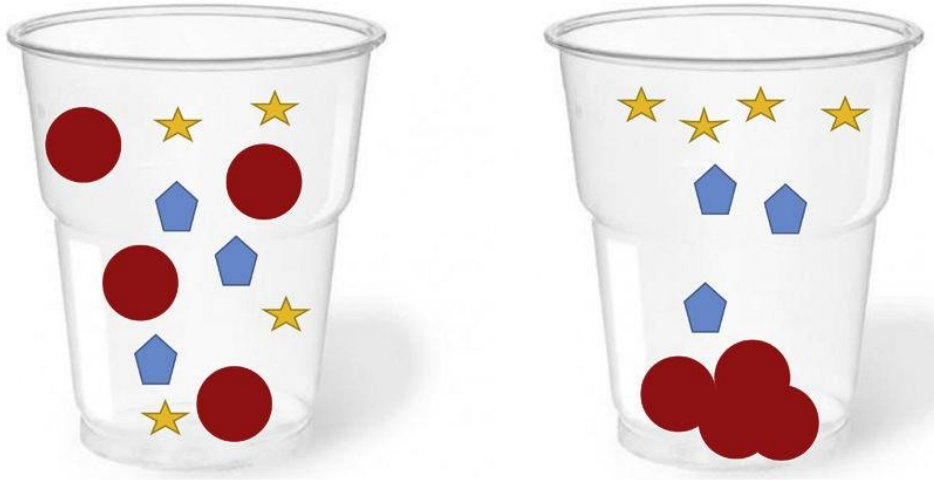
HYPERPROLACTINEMIA SEMIOLOGY





SEXUAL HORMONES DYSFUNCTION+ GALACTHORREA/ GYNECOMASTY

HYPERPROLACTINEMIA INVESTIGATION

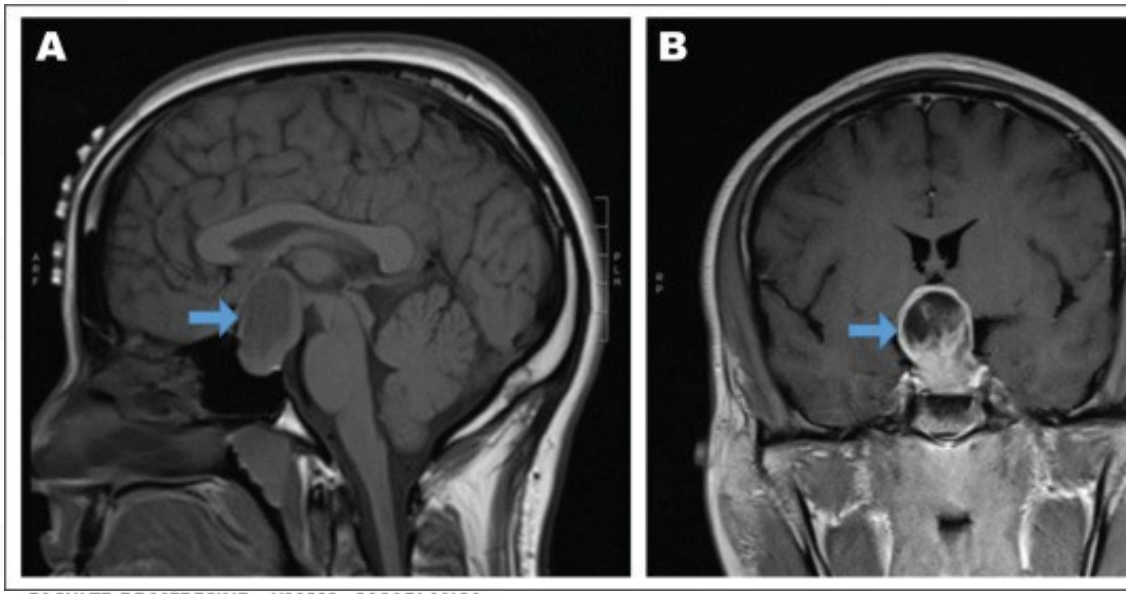


Endocrinol Diabetes Nutr. 2022;69:63–9

Investigations :

- Prolactin assay- POLYETHILENGLICOL (PEG)
-

- MRI of pituitary gland



PROLACTINOMA:
- MICRO
- -MACRO/ INVASIVE

HYPERPROLACTINEMIA ETHYOLOGY

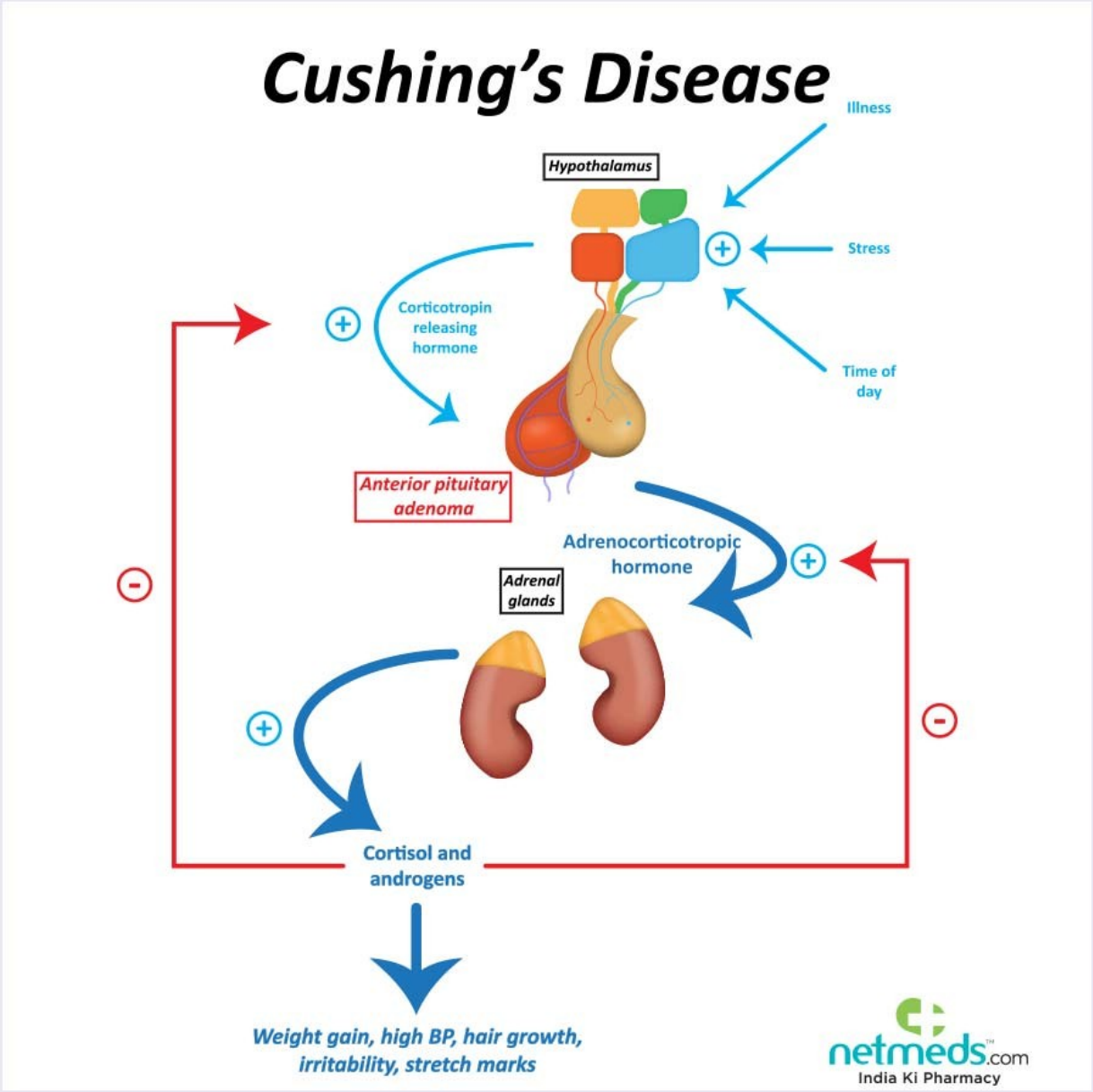
TABLE
Causes of Hyperprolactinemia

Physiologic	Pharmacologic	Pathologic
Pregnancy	Dopamine antagonist (**phenothiazines**[a], haloperidol, **risperidone**[a], **metoclopramide**[a], reserpine, methyldopa, opioids)	Pituitary tumors (prolactinoma, acromegaly that co-secretes growth hormone and prolactin, nonsecretory adenoma)
Nursing	Monoamine oxidase inhibitors	Hypothyroidism
Nipple stimulation	Cimetidine (intravenous)	Chronic renal insufficiency
Exercise	Verapamil	Severe liver failure
Stress	Thyrotropin-releasing hormone stimulation test	Hypophyseal stalk lesion ("stalk effect")
Sleep		Neuraxis irradiation
Seizures		Spinal cord lesions
		Hypophysitis (inflammation of hypophyseal stalk)
		Polycystic ovarian syndrome

[a] Hyperprolactinemia secondary to medications is usually mild (25 to 100 ng/mL), except for the highlighted agents, in which cases levels can exceed 200 ng/mL.

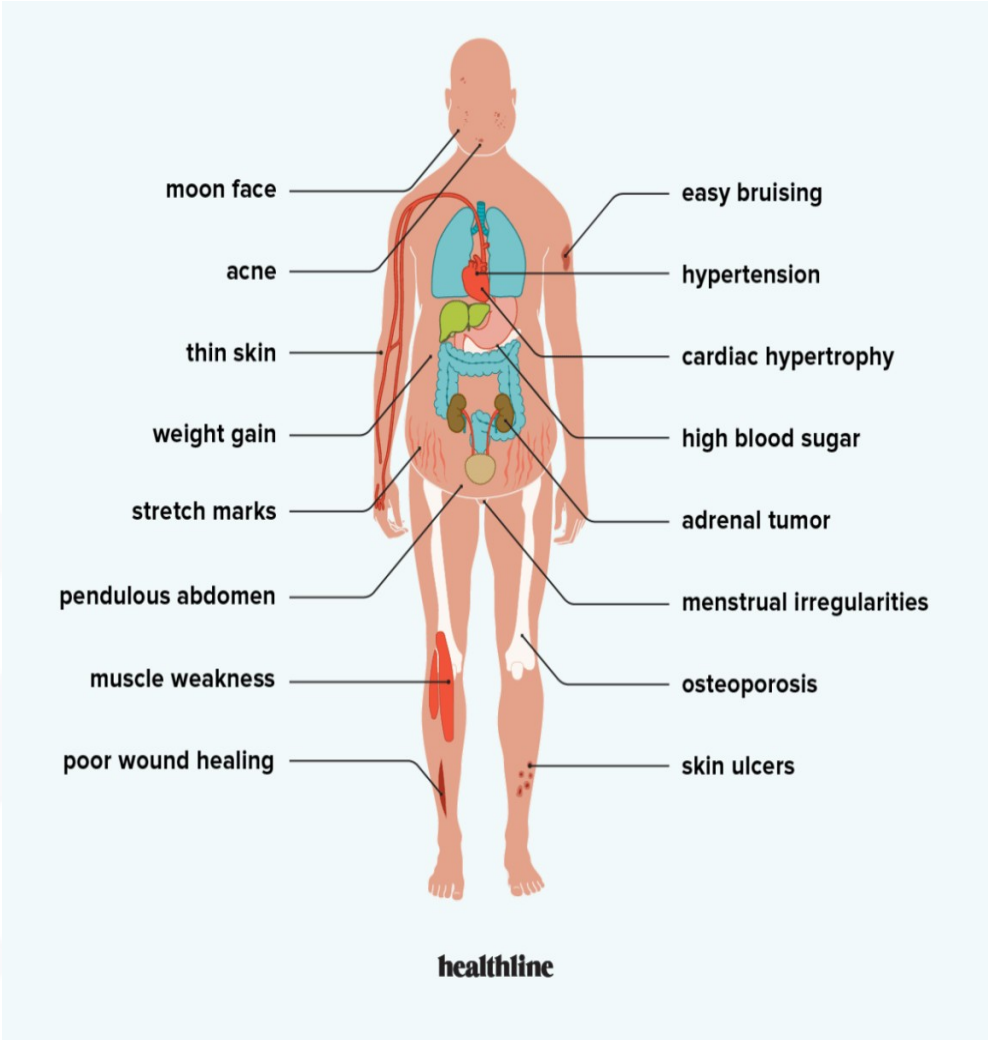
Sources: Gardner and Shoback. 2011.[2] Melmed et al. *J Clin Endocrinol Metab*. 2011.[3]

ACTH HYPERSECRETION- CUSHING DISEASE



CUSHING DISEASE-
ACTH DEPENDENT CORTISOL SECRETION





Cushing disease

Lab tests :

- Free urinary / salivary cortisol
- Cortisol nycthemeral cycle: vesperal cortisol
- Suppressive tests: Dexamethasone 1 mg overnight
- ACTH level

High cortisol level + inadequate supression in DXM test = hypercortisolism

Hypercortisolism + inadequate ACTH= Cushing disease

Pituitary MRI- usually ACTH secreting microadenoma

Clinical Features

1. **Acrofacial Dysmorphism (Disfigurement)**

- **Acromegalic facies:**

✓ Prominence of the supraorbital ridges and frontal bossing;
✓ Enlargement of the nose;
✓ Downward and forward growth of the mandible which leads to prognathism.
✓ Dental problems (teeth separation, malocclusion)
✓ Coarsening of the features,
✓ Deep nasolabial furrows,
✓ Macroglossia and large lips
✓ Sonorous deepening voice



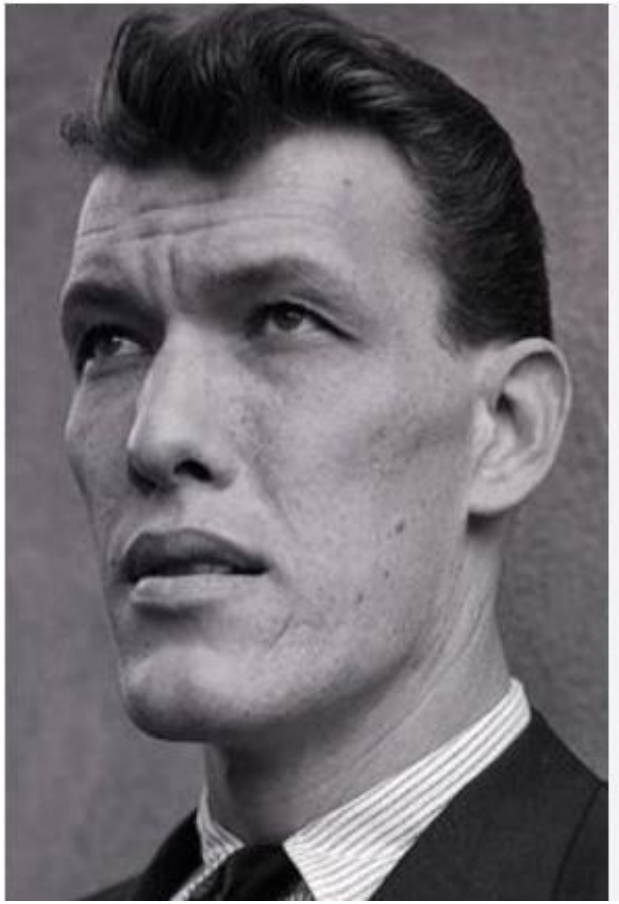
•large, thickened, and bulky hands and feet

- **thickening of the skin, with increased oiliness and sweating.**

Acromegaly







Acromegaly

2. Generalized Visceromegaly

thyromegaly and enlargement of the salivary glands. Enlargement of other organs (liver, heart, kidney)

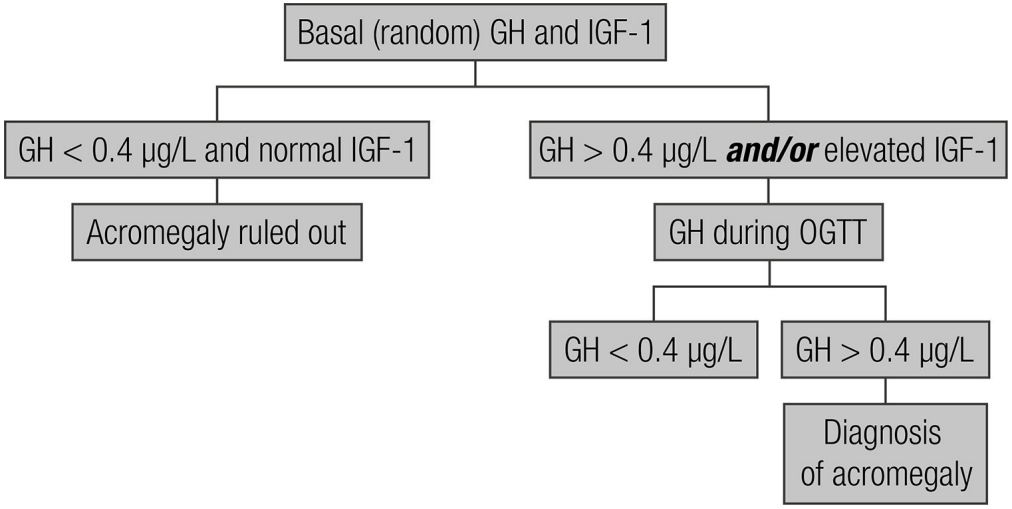
3. Articular and Neural Manifestations =

Acromegalic Arthropathy:degenerative arthritis of the spine, hips, and knees; Neural enlargement and wrist tissue swelling can lead to carpal tunnel syndrome in up to half of all patients characterized by painful median nerve compression and acroparesthesias.

4. **Respiratory Manifestations:** Sleep apnea
5. **Gastrointestinal Features:**colonic polyp
6. **Cardiovascular Manifestations:**Arterial hypertension , Left ventricular hypertrophy
7. **Other Endocrine and Metabolic Abnormalities:**diabetes mellitus/impaired glucose tolerance (IGT), hypertriglyceridemia type IV, hyperlipoproteinemia, hypercalciuria, diffuse or nodular goiter

ACROMEGALY INVESTIGATIONS:

Blood tests



- IGF1
- GROWTH HORMONE (GH)- INHIBITION IN GLUCOSE TOLERANCE 75 G TEST
- PROLACTIN (USUAL CO-SECRETION)

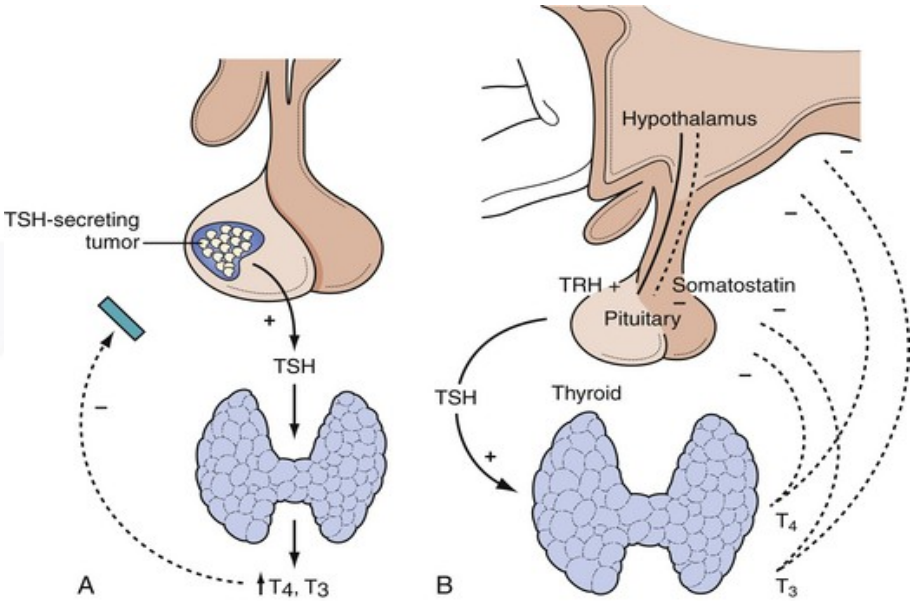
Imagistic:

- Pituitary MRI

Central hyperthyroidism

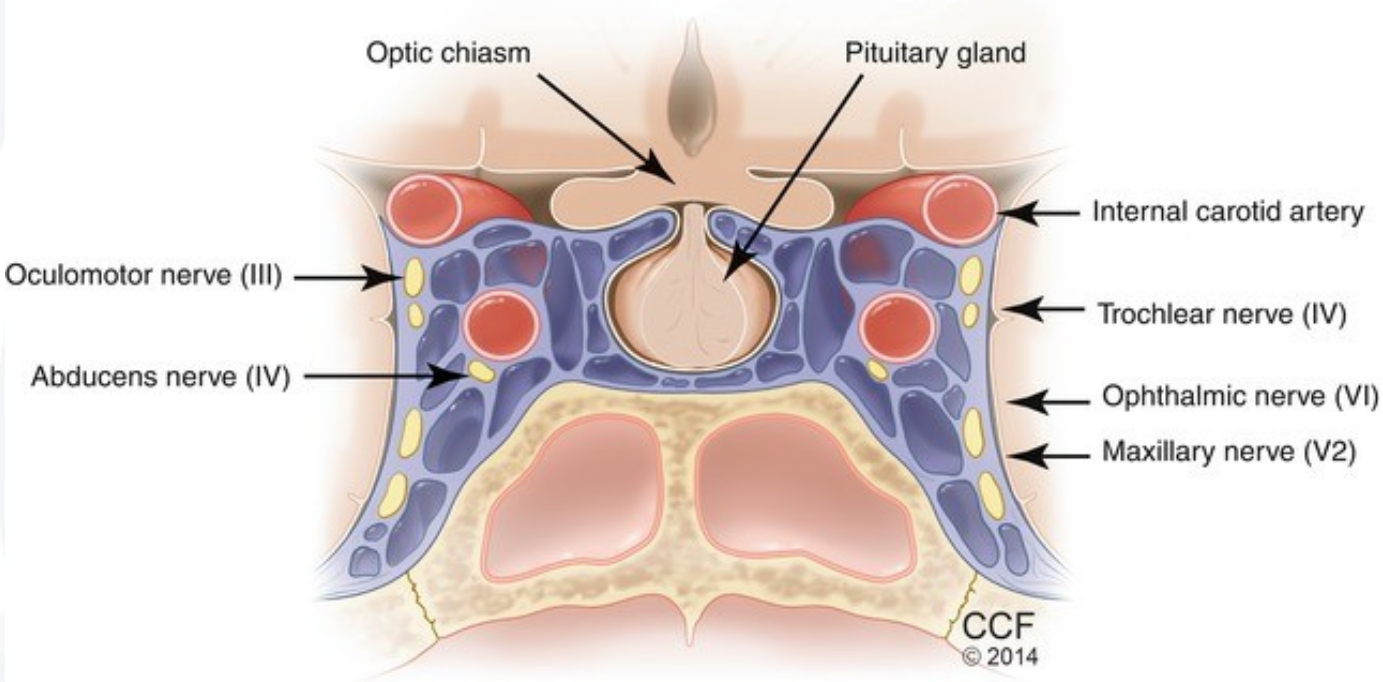
TSH hypersecretion: very rare

- Hyperproduction of TSH
- Pituitary adenoma
- High FT4+FT3 , normale or high TSH



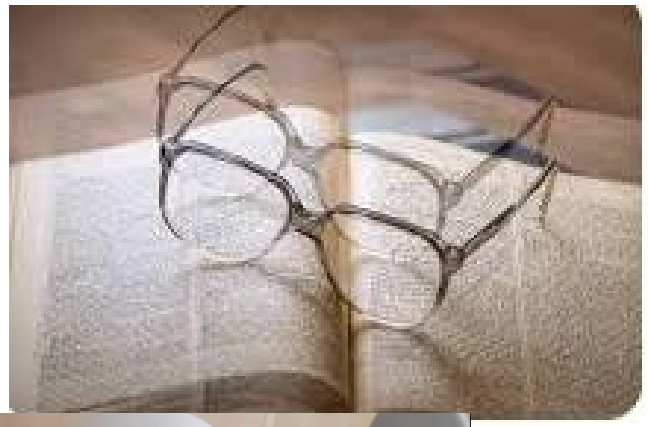
PITUITARY MASS EFFECT

1- VISUAL DYSFUNCTION: DIPLOPIA



COMPRESSION ON CRANIAL NERVES III, IV, VI

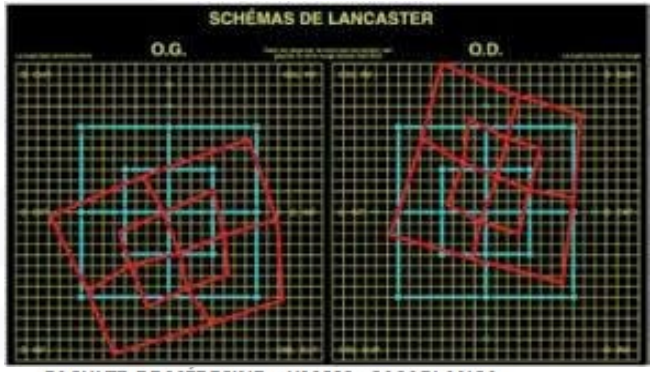
1- VISUAL DYSFUNCTION: DIPLOPIA



2- PALPEBRAL PTOSIS

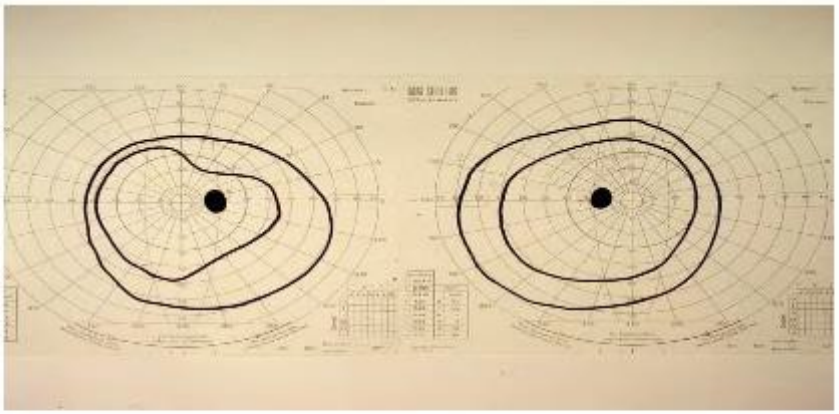


3. Visual field dysfunction

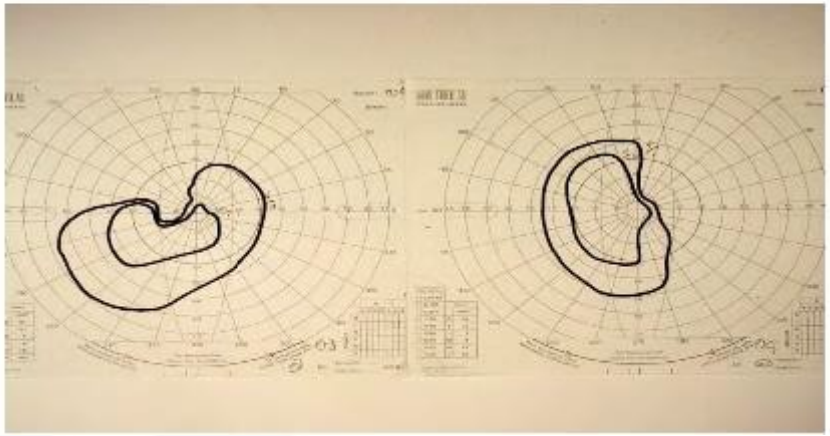


OPTICAL CHIASMA COMPRESSION- VISUAL FIELD AMPUTATION

Normal visuel field



quadrantanopia



homonymous hemianopia

MASS EFFECT

1- 2. Ophthalmological signs:

Suprasellar extension of the pituitary adenoma compresses the optic chiasm leading to the optic—chiasmatic syndrome.:

- **bitemporal visual loss, in the superior field portions = bitemporal sup quadrantanopia bitemporal hemianopia.**,
- **homonymous contralateral hemianopia(the nasal field of the other ocular globe)**
- **Rarely**, decreased visual acuity, pupillary abnormalities, optic atrophy, papilledema, cranial nerve palsies, and nystagmus may be encountered.

PITUITARY APOPLEXY

- **Pituitary apoplexy spontaneous** hemorrhagic infarction of a pituitary tumor (pituitary apoplexy) frequently results in partial or total pituitary insufficiency.
- Fulminant clinical syndrome , a medical emergency.
- Severe, sudden onset headache, visual impairment, ophthalmoplegia (cranial nerve palsy), meningismus, and an altered level of consciousness. Diagnosis is based on the clinical features and pituitary imaging panhypopituitarism
- Early transsphenoidal decompression of the intrasellar contents

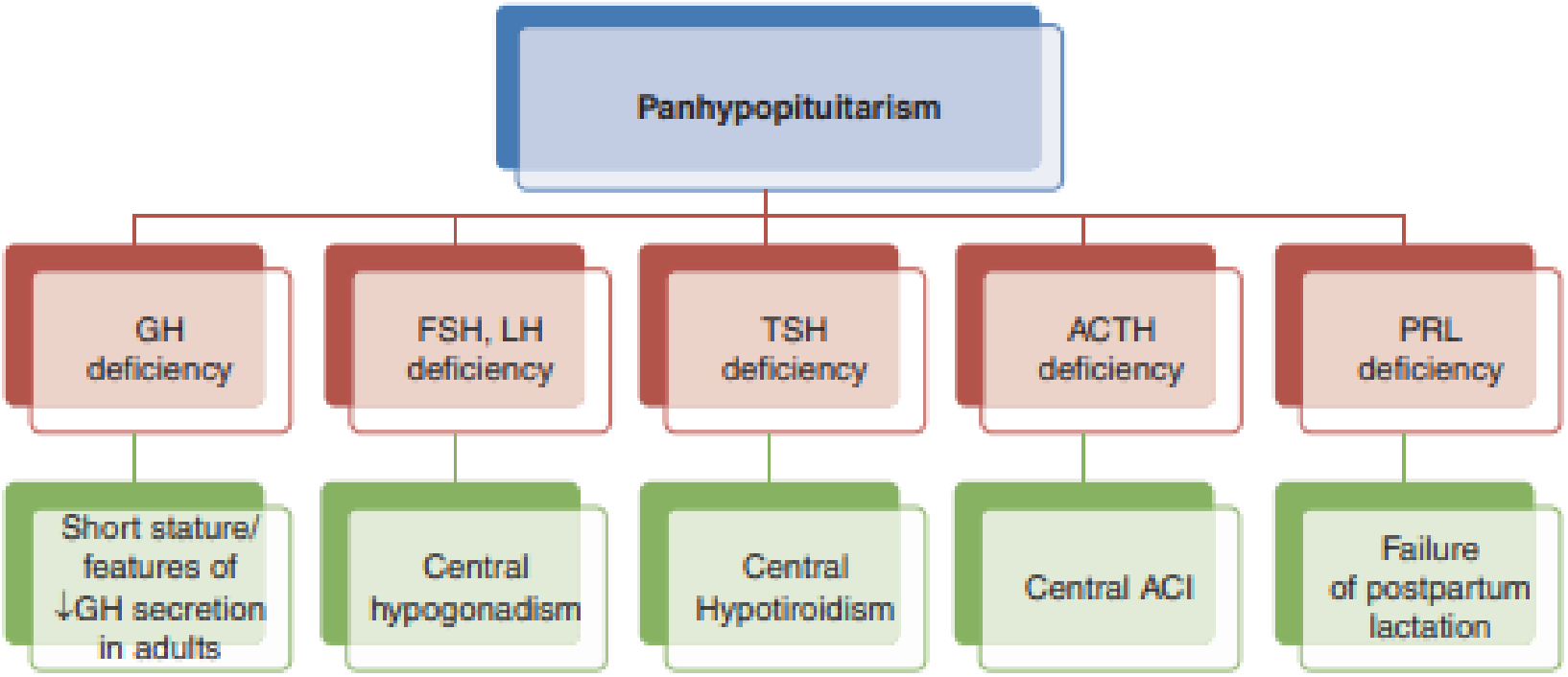
(within 8 days)

Hypopituitarism

Hypopituitarism

- Hypopituitarism is manifested by diminished or absent secretion of **one or more** pituitary hormones (partial or complete deficiency)

Hypopituitarism

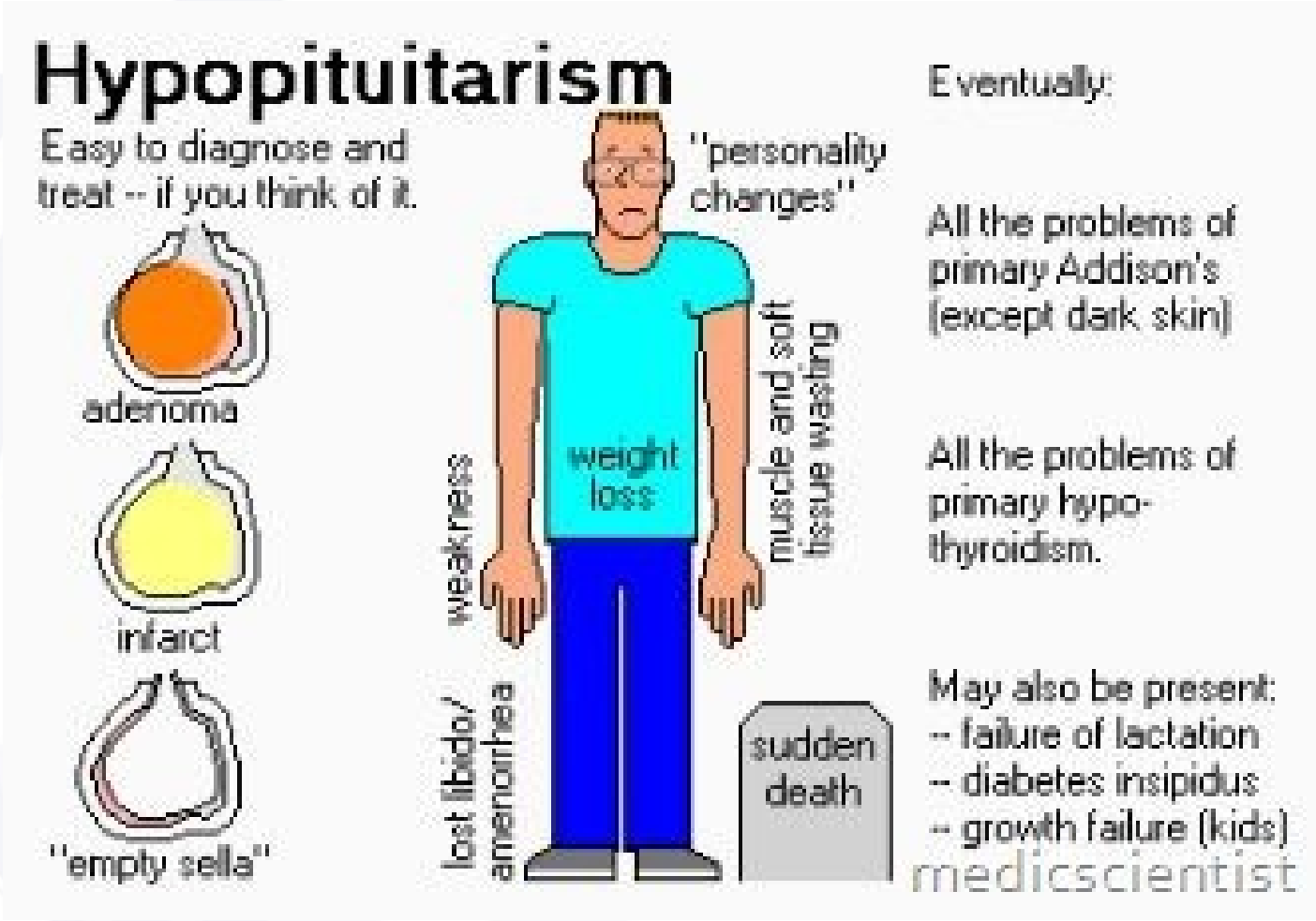


Hypopituitarism

Table 3.2 Main manifestations of hypopituitarism

Hormone deficiency	Clinical features
GH	Children: short stature/adults: reduced exercise capacity, ↓ lean body mass, ↑ body fat, impaired psychological well-being,↑ cardiovascular risk
LH/FSH	• ♀: Oligo/amenorrhea, anovulatory cycles, loss of axillary and pubic hair, osteoporosis • ♂: erectile dysfunction, ↓ libido, testicular atrophy, loss of sexual hair, osteoporosis
TSH	As in primary hypothyroidism, except lack of goiter
ACTH	As in Addison's disease (primary ACI) except lack of hyperpigmentation, hyperkalemia
Prolactin	Failure of lactation
ADH	Polyuria and polydipsia

HYPOPITUITARISM CLINICAL FEATURES



clinical presentation of hypopituitarism

- Pale facies, a bit "old".
- Depigmentation of the mammary areolas and external reproductive organs is constant.
- Thin, cold, dry skin. Wrinkles on the forehead and the outer part of the orbits are fine.
- loss of hair in the axillar and pubic area, due to the absence of adrenal and gonadal androgens.

- **Triad: pallor, depilation, depigmentation**

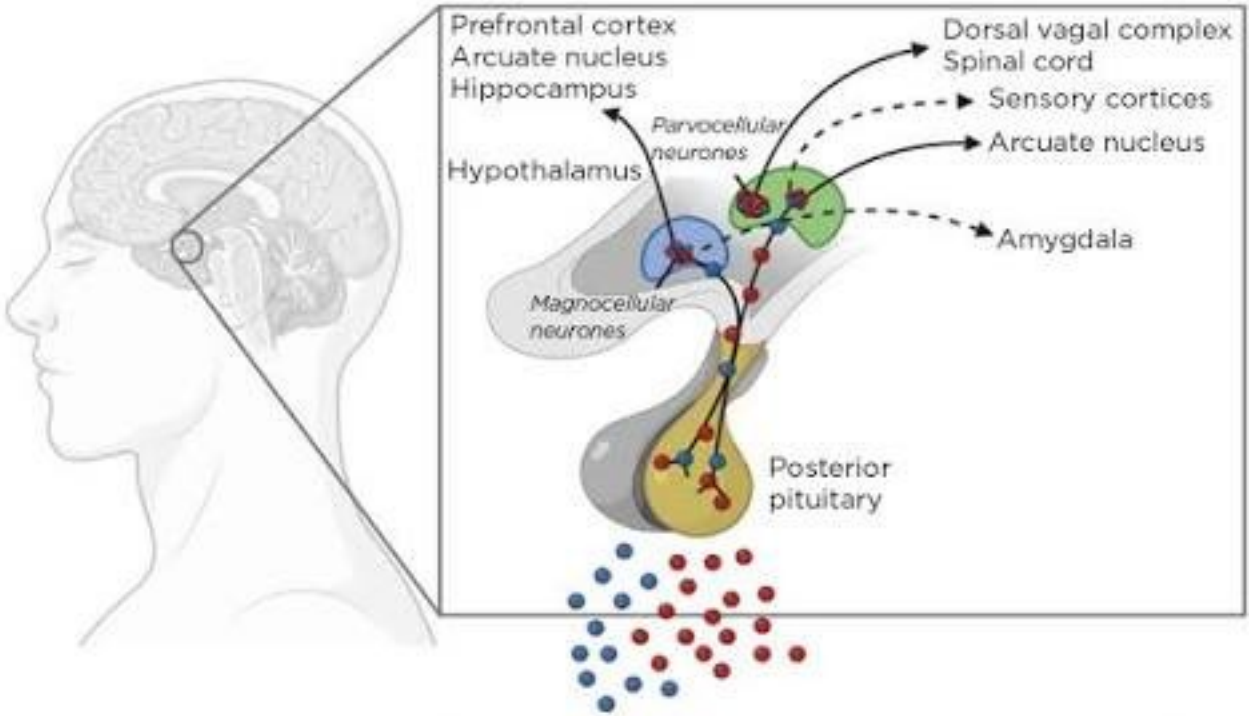


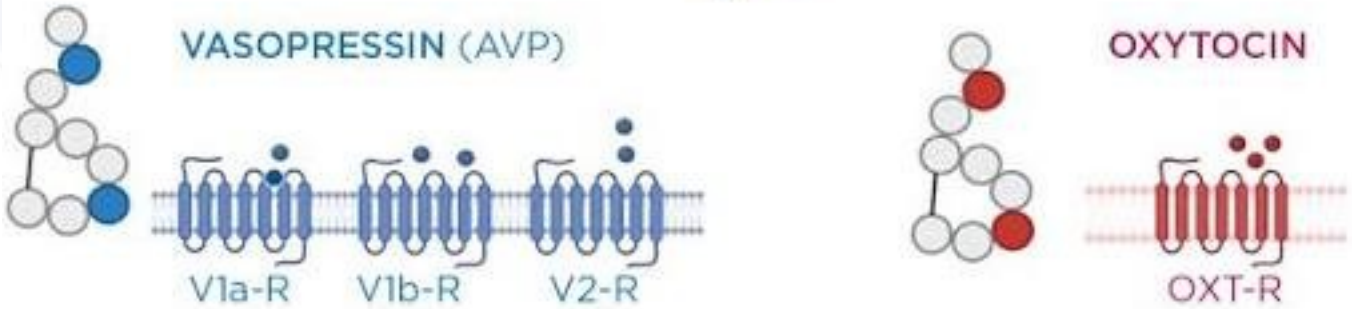


The diagnosis of AHI should lead to pituitary imaging: pituitary MRI

POSTERIOR PITUITARY

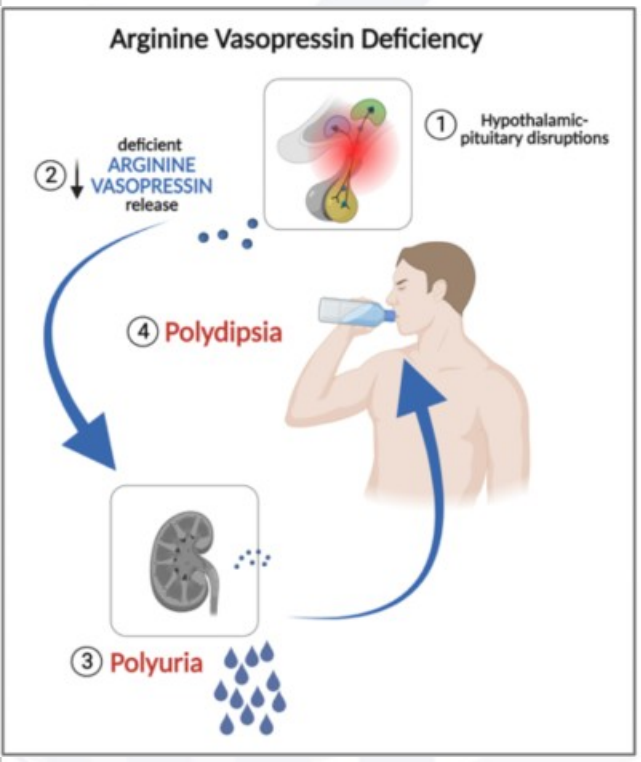
-POSTERIOR PITUITARY HORMONES: DESMOPRESINE (ARGININE VASOPRESSIN) AND OXYTOCINE

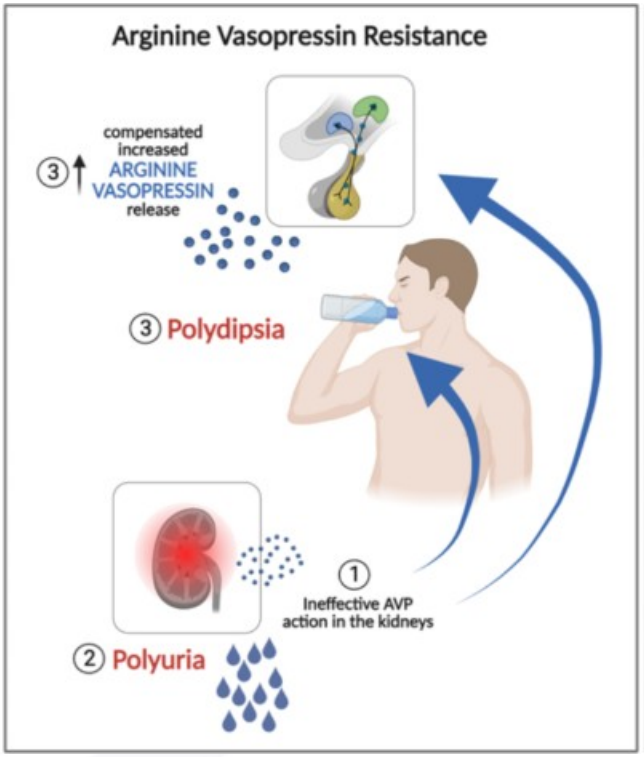


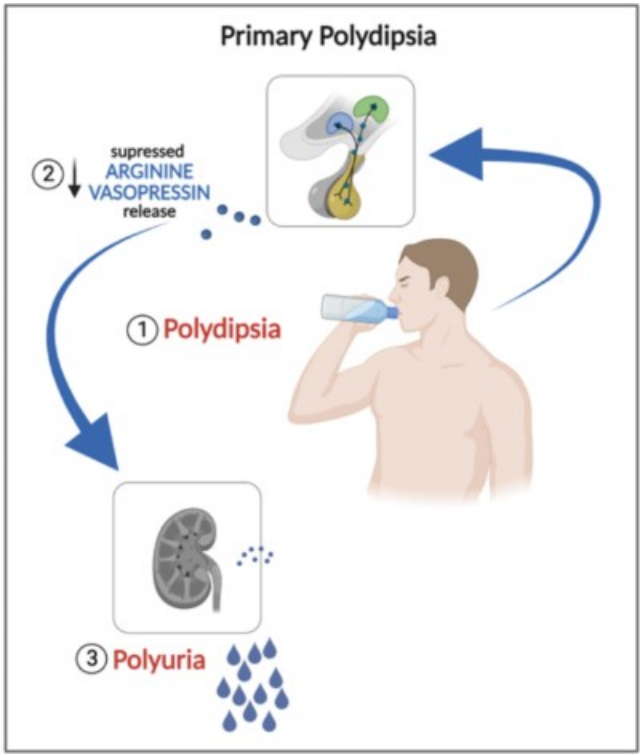


ARGININE VASOPRESSIN DEFICIENCY

DIFFERENTIAL DIAGNOSIS







Ravi Ranjan Kumar

PATIENT CASE - 1111

A 40-year-old man presents with:

- Enlargement of hands and feet
- Increasing shoe and ring size
- Coarse facial features
- Excessive sweating

Examination:

- Frontal bossing
- Prognathism
- Macroglossia



? Question

Diagnosis?

A. Gigantism
B. Acromegaly
C. Hypothyroidism
D. Paget Disease

Ravi Ranjan Kumar
Medico Pharma Experts
For educational purposes only

Cushing clinical manifestations





- Name the clinical manifestations presented:
- A= ?
- B+C?
- D?
- E?

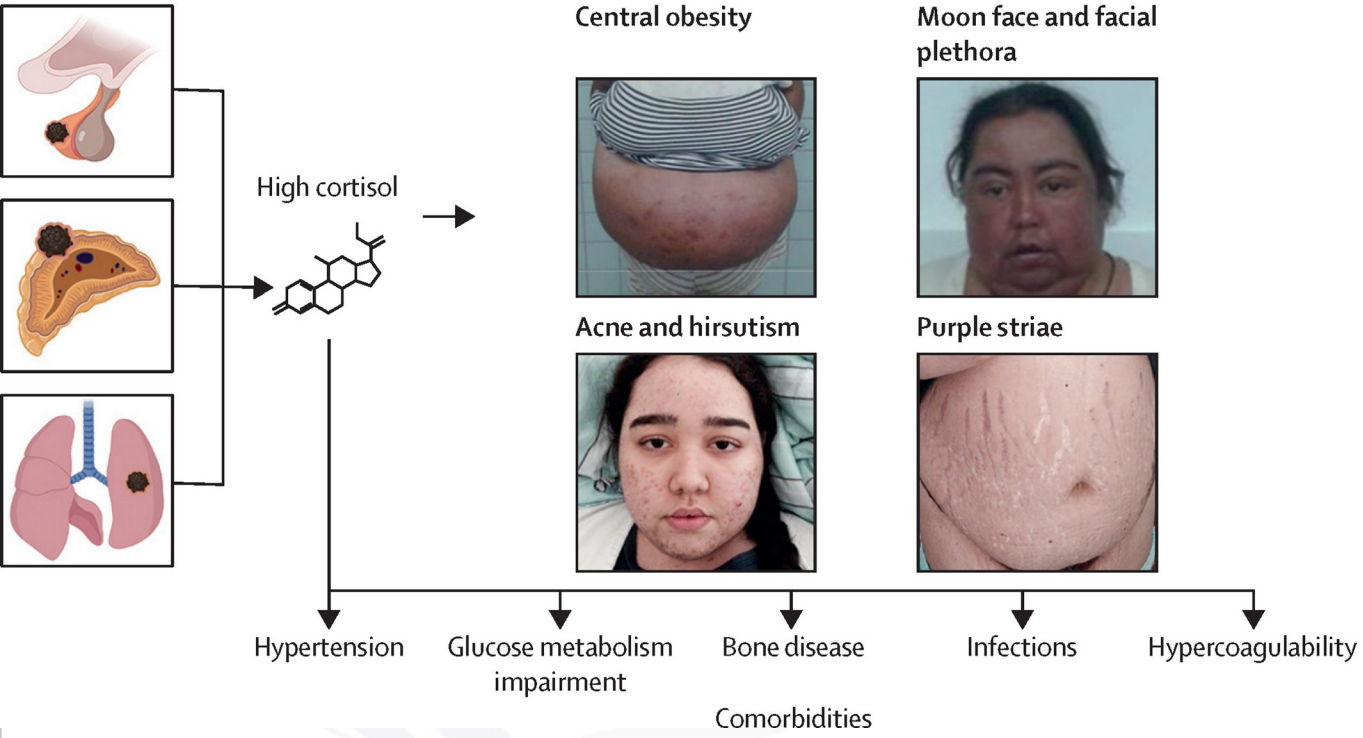
LOOK for Cortisol/ ACTH axis !!

Clinical Case Cushing- Lancet 2023

CORTISOL

Signs
- Central obesity
- Moon face
- Facial plethora
- Acne
- Hirsutism
- Purple striae
- Pubic hair development*
- Growth impairment*

Symptoms
- Decreased libido
- Psychiatric disorders
- Proximal muscle weakness
- Hair loss
- Menstrual irregularities



B

Before transsphenoidal surgery



12 months after transsphenoidal surgery



Volume 402, Issue 10418P2237-2252December 09, Cushing's syndrome
Prof Mônica Gadelha, MD • Federico Gatto, MD[e] • Luiz Eduardo Wildemberg, MD[b] • Prof Maria Fleseriu, MD

Clinical case – post-traumatic hypopituitarisme



Figure

Caption

Figure 1. Forty-nine year-old patient with severe posttraumatic hypopituitarism. Clinical appearance, showing lack of facial hair, dehydration, Queen Anne's sign (panel A), pale skin, muscular atrophy, loss of body hair and anorexia (panel B). T1-weighted coronal MRI of the pituitary region, with flattened pituitary gland at the bottom of the sella (panel C, yellow arrow) and contusional defects in the temporal lobe (panel D, white arrow). Tissue adjacent to the right cavernous sinus (panel C, red arrow) compatible with dislocated pituitary tissue after fracture of the middle cranial fossa.

What to expect on anterior pituitary hormonal evaluation?

1. Cortisol level?
2. TSH/ fT4?
3. GH/ Prolactin?



PITUITARY & FRIENDS

